

PROJECT DOCUMENTATION

ANTERIOR CHAMBER DEPTH: CORNEAL THICKNESS

INTRODUCTION

Glaucoma is a condition that causes damage to your eye's optic nerve and gets worse over time. It is often linked to a buildup of pressure inside your eye. Glaucoma tends to be inherited and may not show up until later in life.

The increased pressure, called intraocular pressure, can damage the optic nerve, which transmits images to your brain. If the damage continues, glaucoma can lead to permanent vision loss. Without treatment, glaucoma can cause total permanent blindness within a few years.

Most people with glaucoma have no early symptoms or pain. You need to see your eye doctor regularly so she can diagnose and treat glaucoma before long-term visual loss happens.

If you are over age 40 and have a family history of the disease, you should get a complete eye exam from an eye doctor every 1 to 2 years. If you have health problems like diabetes or a family history of glaucoma or are at risk for other eye diseases, you may need to go more often.

Cause of glaucoma is high fluid pressure inside your eye. This happens when the liquid in the front part of the eye does not circulate the way it should.

Normally, the fluid, called aqueous humor, flows out of your eye through a mesh-like channel. If this channel is blocked, the liquid builds up. That is what causes glaucoma. The reason for the blockage is unknown, but doctors do know it can be inherited, meaning it is passed from parents to children.

Less common causes include a blunt or chemical injury to your eye, severe eye infection, blocked blood vessels inside the eye, and inflammatory conditions. It is rare, but sometimes eye surgery to correct another condition can bring it on. It usually affects both eyes, but it may be worse in one than the other.

There are two main kinds:

Open-angle glaucoma. It is the most common type also called as wide-angle glaucoma. The drain structure in your eye -- it is called the trabecular meshwork -- looks normal, but fluid does not flow out, as it should.

Angle-closure glaucoma. It is less common in the West than in Asia. You may also hear it called acute or chronic angle-closure or narrow-angle glaucoma. Your eye does not drain right because the angle between your iris and cornea is too narrow. Your iris is in the way. This can cause a sudden buildup of pressure in your eye. It is also linked to farsightedness and cataracts, a clouding of the lens inside your eye.

Glaucoma causes loss of vision if not treated at the initial stage. Regular eye examinations by an ophthalmologist that includes eye pressure measurement (tonometry); inspection of drainage angle of the eye (gonioscopy); evaluation of optic nerves (ophthalmoscopy); and testing the visual field of the eye (perimetry).

Thus, the regular check-up for glaucoma is an expensive affair. Thus, finding an alternative cheaper way is the main aim of the project. The project also focuses on making the device compact and mobile.

VAN HERICKS TECHNIQUE

The **Van Herick technique** is an eye examination method used to determine the size of the anterior chamber angle of the eye.

Glaucoma is currently the principal cause of irreversible blindness in the world. Therefore, The Van Herick technique is pertinent to the interests of all eye care practitioners as it permits for a quick and simple screening alternative to the conventional Gonioscopy examination.

The Van Herick's technique has become the most commonly used qualitative method of assessing the size of the anterior chamber angle (ACA). Whereby, it involves comparing the depth of the peripheral anterior chamber to the thickness of the cornea, when a narrow beam is shone within the limbus at a 60° angle. The anterior chamber drainage angle is then graded as a ratio between the peripheral anterior chamber depth and corneal thickness (AC : C ratio) or expressed traditionally as a fraction to provide the Van Herick's result. Grading can also be obtained by distinguishing the structures visible upon observation. Each visible structure correlates to a Van Herick grade which can then be expressed a fraction.

METHOD

The technique was initially designed by Van Herick, performed with the utilization of a slit lamp without the requirement of any additional aids. It facilitates a simple and quick assessment of the lateral chamber angle. The method involves a narrow slit of light from a slit lamp being projected onto the peripheral cornea at an angle of 60° as near as possible to the limbus. The resulting image is a slit that is projected onto the surface of the cornea, the width is then used as a reference for the grading of the angle. The width of the angle is graded by the distance between the corneal slit image and the slit image on the iris.

Chamber Depth in Corneal Thickness	Angle Grade
$\geq \text{cornea}$	4
$1/4 \text{ to } 1/2$	3
$1/4$	2
$<1/4$	1
Slit	Dangerously Narrow

From the above figure, the severity decreases from grade 0 to grade 4.

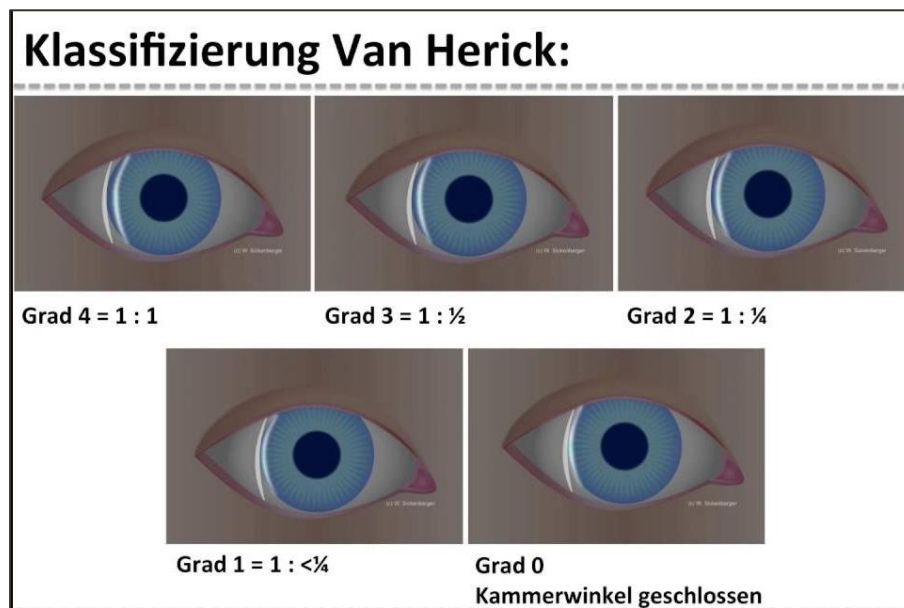


Fig. Van Herick technique gradation

PROJECT DESCRIPTION

The ACDC prototype is two-armed system with the arms at angle of 60 degree. One arm consists of the slit lamp and other the camera to capture the image. The device is to be placed at distance of approx. 5cm from the eye such that camera contained arm is at 90 degree to the eye. Thus the slit lamp ray falls at an angle of 60 degree to the eye, satisfying van Herick theory.

Slit lamp consists of three lenses (two lenses of 20D and one of 90D). Slit is of 0.1 mm width, as the requirement is of very thin beam of light so that the reflections are well distinguished. The position of the lenses is maintained such that the final beam of light is parallel and sharp.

The first 20D lens converts the diverged beam of light from the LED to parallel beam(as LED is at the focal length of the lens). The second 20D lens is kept to encounter the effect of first. The parallel beam is converted to converging beam of light. The slit is at a distance less than the focal length. The concentrated beam of light passes through the slit of 0.1 mm. The 90D lens converts the thin line of beam into parallel beam, to maintain the shape of varied range of distance.

The camera used is piCamera (raspberry pi 3).

Led- intensity less than 900lux



Fig. Lens alignment of the prototype



Fig. ACDC Device

IMAGE PROCESSING

The images captured using the camera are fed to raspberry pi, which is coded to process the image that calculates the ratio of the anterior chamber depth and corneal distance.

Image processing consists of conversion the RGB image into gray scale image, binarisation of grayscale image using Otsu's algorithm, contour detection, and width measuring algorithm.

Gray scale conversion: converts RGB image to gray scale image to get the intensity pixel values.

Otsu's binarisation: Otsu's algorithm is used to find the optimum threshold for binarisation. This algorithm works only for bimodal images (i.e. which have two prominent peaks in the

histogram). It plots the histogram of the gray scale image and finds the pixel value such that the variance around both the peaks in the histogram is minimum. Using the algorithm the reflection part of the image can be properly distinguished as the intensity of that part is high as compared to the other pixels.

Contour detection: Contours can be explained simply as a curve joining all the continuous points (along the boundary), having same color or intensity. The contours are a useful tool for shape analysis and object detection and recognition. This algorithm works better for binary images.

Width measurement: A loop is run that looks for the change in the pixel value from black to white. With this loop we get three value of pixel from the edges of the two reflections. Subtracting these three values get the thickness value of corneal thickness and anterior chamber depth. Calculating the ratio gives us the final expected output.

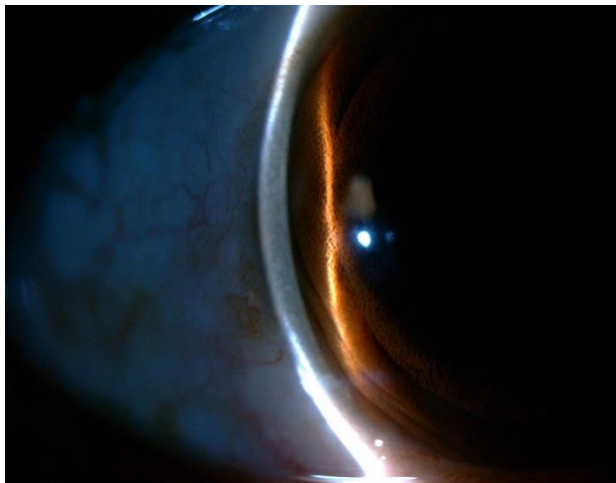


Fig. RGB image from the slit lamp



Fig. cropped and otsu's binarised image

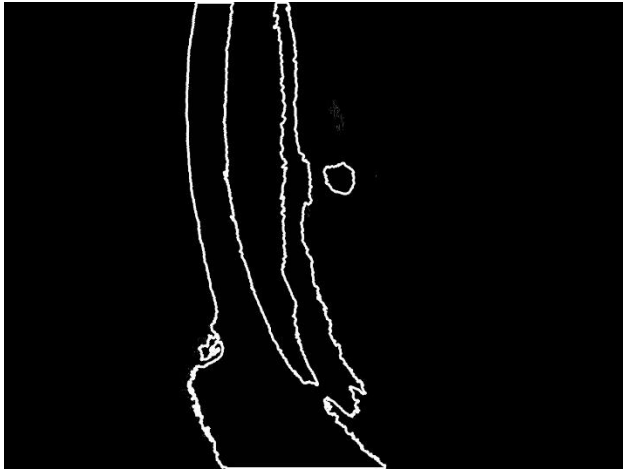


Fig. contour detected image